

[illegible]

© Information Technology Department | 2025



TABLE OF CONTENTS

1. PURPOSE	5
2. OVERVIEW	5
3. AUTHENTICATION & AUTHORIZATION	6
3.1. AUTHENTICATION FLOW	6
3.2. SCOPES & POLICIES	6
3.3. REFRESH TOKEN	6
RATE LIMITS & THROTTLING	7
4. COMMON REQUEST / RESPONSE CONVENTIONS	7
5. STATUS CODES & ERROR FORMAT	7
5.1.1 Common HTTP codes used	7
5.1.2 Standard error object	8
6. ENDPOINTS (DETAILED)	9
6.1. AUTHENTICATION	9
6.2. DATA LIBRARY	10
6.2.1 GET /api/v1/datalibrary/frequencies	10
6.2.2 GET /api/v1/datalibrary/sectors	11
6.2.3 GET /api/v1/datalibrary/subjects	11
6.2.4 GET /api/v1/datalibrary/subjects/sector/{sectorCode}	11
6.2.5 GET /api/v1/datalibrary/datacodelist	11
6.2.6 GET /api/v1/datalibrary/datavalues	12
6.2.7 GET /api/v1/datalibrary/exchangerates	12
6.3. ECONOMIC INDICATORS	14
6.3.1 Weekly Economic Indicators	14
6.3.2 Monthly Economic Indicators	14
7. EXAMPLE REQUESTS	14
7.1. GET TOKEN (CURL)	14
7.2. CALL PROTECTED ENDPOINT	14
7.3. POSTMAN: PASSING LIST<INT> FOR GET QUERY	15
VERSIONING & CHANGELOG	16
8. CONTACT & SUPPORT	16



Amendment History

Date	Version	Description	Author



Document Approval

Department of Information Technology, Central Bank of Sri Lanka (CBSL) together with Finance Department, Domestic Operations Department and Currency Department of the CBSL have reviewed this document and hereby agree that the contents herein are accurate.

Any changes to this document must be communicated in writing and signed off by the respective parties.

User Department	Name	Designation	Signature	Date
Finance				
Domestic Operations				
Currency				

Information Technology Department	
Name	
Designation	
Signature	
Date	



1. Purpose

This API provides a secure, reliable interface that allows external developers, businesses, and the public to access Central Bank (CBSL) data and services programmatically.

Main objectives:

- Provide secure, reliable, and near-real-time access to CBSL data.
- Enable reuse of data in analytics, reporting, research and 3rd-party services.
- Provide a foundation for future digital initiatives and integration (CBDC experiments, open banking compatibility).

2. Overview

Base URL (example):

<https://localhost:7091>

API supports JSON responses. Authentication uses OAuth2 client-credentials (Bearer JWT). Tokens issued by [/connect/token](#).



3. Authentication & Authorization

3.1. Authentication flow

This API uses **OAuth2 Client Credentials** to issue JWT bearer tokens.

Token endpoint

POST /connect/token

Content-Type: application/x-www-form-urlencoded

```
grant_type=client_credentials  
client_id={clientId}  
client_secret={clientSecret}
```

Success response (200 OK):

```
{  
  "access_token": "eyJhbGciOi...",  
  "token_type": "Bearer",  
  "expires_in": 3600,  
  "scope": "common account statistics",  
  "refresh_token": "....",  
  "refresh_token_expires_in": 2592000  
}
```

How to use the token

Add HTTP header to API calls:

Authorization: Bearer {access_token}

3.2. Scopes & policies

The API defines scopes and authorization policies that map to claims in the token:

- common — general access data
- account — access to account / data code list / data values
- statistics — access to statistics endpoints

Controller actions use [Authorize(Policy = "account")], [Authorize(Policy = "statistics")], etc.

Clients must include scope claims matching policy requirements. Tokens issued by /connect/token contain scope claim values (one claim per scope).

3.3. Refresh token

Endpoint: POST /connect/refresh/token

Used to exchange refresh token for a new access token (grant_type = refresh_token with client_id, refresh_token).



Rate Limits & Throttling

(If you have operational limits, list them here. Example recommended policy:)

- Default: **100 requests per second per client.**
- Burst: allow short bursts up to **1,000 requests.**
- If exceeded: 99 Too Many Requests, with Retry-After header in seconds.

Add logging/auditing of heavy clients; require API usage agreement for higher limits.

4. Common Request / Response conventions

- All requests and responses use application/json unless otherwise specified.
- Dates format: yyyy-MM-dd (ISO date). Where time is returned, use ISO 8601 (UTC) yyyy-MM-ddTHH:mm:ssZ.
- Decimal numbers: JSON numbers (no thousands separators). When parsing from DB, transforms strip commas.
- Paging: Use query params page and pageSize where result sets may be large.
- Errors always return structured JSON (see Status Codes & Error Format).

5. Status Codes & Error Format

5.1.1 Common HTTP codes used

- 200 OK — Successful response. Body contains requested data.
- 201 Created — Resource created (not used commonly for read API).
- 204 No Content — Success with no body (e.g., delete).
- 400 Bad Request — Input validation failed.
- 401 Unauthorized — Missing/invalid authentication.
- 403 Forbidden — Authenticated but insufficient permission (scope).
- 404 Not Found — Resource not found.
- 409 Conflict — Data conflict.
- 422 Unprocessable Entity — Business validation failure.
- 429 Too Many Requests — Rate limit exceeded.



- 500 Internal Server Error — Unexpected server error.

5.1.2 Standard error object

All errors return an RFC-like JSON body:

```
{  
  "error": "invalid_request",  
  "error_description": "fromDate cannot be greater than toDate",  
  "details": null,  
  "timestamp": "2025-10-24T10:00:00Z"  
}
```

- error is a short machine error code.
- error_description is a human message.
- details may contain validation detail objects.
- timestamp for troubleshooting.



6. Endpoints (detailed)

All endpoints shown with method, path, auth, query/body, and sample responses.

6.1. Authentication

POST [/connect/token](#)

- Auth: **No** (public)
- Content-Type: application/x-www-form-urlencoded
- Body:
 - grant_type: client_credentials
 - client_id
 - client_secret
- Response: access token JSON (see earlier).

POST [/connect/refresh/token](#)

- Auth: **No** (public)
- Body:
 - grant_type: refresh_token
 - client_id
 - refresh_token
- Response: new access token JSON.



6.2. Data Library

6.2.1 GET /api/v1/datalibrary/frequencies

- Auth: Required
- Query: none
- Response 200 OK:

```
[  
  { "code": "M", "name": "Monthly" },  
  { "code": "Q", "name": "Quarterly" }  
]
```



6.2.2 GET /api/v1/datalibrary/sectors

- Auth: **AllowAnonymous**
- Response 200 OK:

```
[  
  { "sectorCode": "AGR", "sectorName": "Agriculture" },  
  ...  
]
```

6.2.3 GET /api/v1/datalibrary/subjects

- Auth: **Required**
- Response 200 OK:

```
[  
  { "id": 12, "name": "National Accounts", "sectorCode": "ECO" },  
  ...  
]
```

6.2.4 GET /api/v1/datalibrary/subjects/sector/{sectorCode}

- Auth: client1 policy required.
- Path param: sectorCode (string)
- Response 200 OK: filtered subjects.

6.2.5 GET /api/v1/datalibrary/datacodelist

- Auth: account policy
- Query:
 - subId (int?, optional)
 - sectorCode (string?, optional)
 - frequencyCode (string?, optional)
- Response 200 OK:

```
[  
  {  
    "dataCodeId": 123,  
    "itemName": "Subject - Topic",  
    "shortName": "GDP",  
    "unit": "LKR",  
    "scale": "1",  
  }  
]
```



```
"frequencyCode": "Q"
},
...
]
```

6.2.6 GET /api/v1/datalibrary/datavalues

- Auth: account policy
- Query:
 - dataCodeList: repeated param for list of ints (e.g. ?dataCodeList=982&dataCodeList=999)
 - notes (bool)
 - from (string date yyyy-MM-dd)
 - to (string date)
 - frequencyCode (string)
- Response 200 OK: DataValueReturn object:

```
{
  "dataValueMain": [ /* DataValueMain[] for each requested DataCode */ ],
  "dataValueNotes": [ /* DataValueNotes[] if requested */ ]
}
DataValueMain:
{
  "itemName": "Subject - Topic",
  "dataValues": [
    { "period": "2017-02-28", "value": "237.847/234.2258", "unit": "LKR", "scale": "1" },
    ...
  ]
}
```

Notes

- For long lists, prefer POST with JSON (see Best Practices).

6.2.7 GET /api/v1/datalibrary/exchangerates

- Auth: optional (depends on your policy)
- Query:
 - currencyCode (string, e.g. USD)
 - typeLike (string e.g. ex-bs, ex-ad)



- fromDate (string, yyyy-MM-dd)
- toDate (optional)
- Validation: fromDate and toDate must parse to DateTime and fromDate <= toDate.
- Response 200 OK (when typeLike=ex-bs return ExchangeRateBuyingSelling[]):

```
[  
  {  
    "date": "2021-01-08",  
    "type": "Exchange Rates - Buying / Selling (USD) - Rs",  
    "currencyCode": "USD",  
    "buyRate_LKR": 186.28,  
    "sellRate_LKR": 190.98  
  }  
]
```



6.3. Economic Indicators

6.3.1 Weekly Economic Indicators

6.3.1.1. Price Indices

- Auth: No auth
- Query:
 - period (string, yyyy-MM-dd)
 - format (string, optional)
- Validation: fromDate and toDate must parse to DateTime and fromDate <= toDate.
- Response 200 OK (when typeLike=ex-bs return ExchangeRateBuyingSelling[]):

```
[  
  {  
    "date": "2021-01-08",  
    "type": "Exchange Rates - Buying / Selling (USD) - Rs",  
    "currencyCode": "USD",  
    "buyRate_LKR": 186.28,  
    "sellRate_LKR": 190.98  
  }  
]
```

6.3.2 Monthly Economic Indicators

7. Example Requests

7.1. Get token (curl)

```
curl -X POST https://localhost:5001/connect/token \  
-d "grant_type=client_credentials" \  
-d "client_id=client1" \  
-d "client_secret=secretA123!" \  
-d "scope=account statistics"
```

7.2. Call protected endpoint

```
curl -H "Authorization: Bearer eyJhbGciOi..." \  

```

```
"https://localhost:5001/api/v1/datalibrary/datavalues?dataCodeList=982&dataCodeList=99  
9&notes=true&from=2015-01-01&to=2017-12-31"
```



7.3. Postman: passing List<int> for GET query

In Postman Params tab provide multiple dataCodeList keys:

dataCodeList=982

dataCodeList=999

notes=true

from=2015-01-01

to=2017-12-31

Or use POST /api/v1/datalibrary/datavalues/query with JSON body for heavy queries (recommended).



Versioning & Changelog

- Document version 1.0 — date 2025-10-24.
- Add breaking changes with new minor/major versions and maintain /api/v1/ prefix.
- Example changelog entry format:

v1.0 (2025-10-24) - Initial release: frequencies, sectors, subjects, datacodelist, datavalues, exchangerates, auth token endpoints.

8. Contact & Support

- API Owner: OpenAPI Team / CBSL Integration Team
- Contact: api-support@cbsl.lk
- For production issues: open a ticket with correlation id and request details including timestamps.